

Dissimilarity can promote social learning

Abstract

Ample research shows that human social learning is subject to a similarity bias: People are more likely to copy the behavior of those who resemble themselves. Here we hypothesize that in situations where different approaches must be combined to improve a solution, individuals may instead be more open to learning from dissimilar others because they expect valuable solutions from them. We test this expectation in a preregistered experiment where 859 individuals first developed suboptimal approaches to a problem-solving task and were subsequently exposed to the behavior of a demonstrator. The demonstrator pursued an alternative suboptimal approach; and participants could learn to optimize their own approach by combining it with what they observed. A 2x2 experimental design varied whether the demonstrator had shared or dissimilar characteristics, and whether (dis-)similarity related to ideological leaning or the outcome of a fictional cognitive style test. Results show that participants optimized more often after exposure to someone with a different versus similar cognitive style. Ideological (dis-)similarity, on the other hand, did not affect optimization. Our study shows that characteristics pointing to differences in perspectives can promote social learning, challenging the common assumption that people learn best from those similar to themselves.

Keywords: social learning | similarity bias | optimization | learning traps

1 **Introduction**

2 Humans rely heavily on social learning to navigate complex ecological and social environments
3 [1–3]. From early childhood onward, individuals acquire a wide range of knowledge and skills—
4 about norms, language, cooking, parenting, and other everyday practices—by observing others
5 rather than by discovering everything on their own [4]. Because not all information is equally
6 useful, successful social learning depends on strategies that specify who to learn from, what to
7 copy, and when to rely on others [5]. Effective social learning is crucial not only for individual
8 success, but also the functioning of larger collectives and their ability to respond to problems [6],
9 to innovate [7], and to adapt to changing environments [8].

10 A well-documented learning strategy of individuals is to preferentially learn from similar others
11 [9]. Across childhood and adulthood, people tend to copy individuals who share salient
12 characteristics with them—such as age, gender, or ethnicity—on the assumption that similarity
13 makes others’ behavior more relevant to them [10]. Most of the existing research, however,
14 examines cases of social learning where individuals simply reproduce existing behavior. Kinzler
15 [11], for example, shows that children are more likely to adopt languages from adults with the
16 same native dialect as their parents, and Chartrand & Lakin [12] show that infants tend to mimic
17 gestures from same-gender peers. Similarly, Reyes-García and colleagues [13] indicate that
18 adults preferentially copy cultural practices from people with similar socio-demographic traits.

19 This perspective overlooks situations where social learning is most beneficial when one
20 integrates, rather than reproduces, someone else’s approach. In such cases, copying information
21 wholesale will not lead to success; rather, its value only emerges from carefully combining
22 different approaches. Individual improvement therefore depends on the learners’ willingness to
23 engage with different perspectives or practices and to explore whether aspects of others’

approaches can be integrated to improve their own approach. In this context, whether individuals are more likely to learn from others who are similar to them or from those who are dissimilar remains largely unknown.

On the one hand, and as a large literature on social identity and associated biases in human groups shows, dissimilarity might trigger outgroup devaluation [14], leading individuals to view dissimilar others as less competent and to disregard their behavior. Research on the spread of novel behaviors, such as the adoption of new tools and techniques [15], health behaviors [16], or consumption patterns [17], shows that peer dissimilarity decreases the likelihood of adoption. Conversely, social information is more likely to be considered when it comes from ingroup sources [18], and adoption from ingroup members becomes more likely the stronger individuals identify with their own group [19].

On the other hand, in situations where different approaches must be combined to improve a solution, social learners may preferentially seek out dissimilar individuals because they expect them to hold valuable perspectives that they themselves and similar others do not have access to. This conjecture is supported by experimental research on group decision-making, showing that demographic diversity can increase individuals' openness to unfamiliar information [20–23]. In these experiments, dissimilar demographics legitimized dissent and led individuals to take approaches from dissimilar others more seriously. Conversely, unfamiliar information from similar individuals was ignored more frequently and regarded as implausible. These findings suggest that dissimilarity might encourage learners to engage with and integrate perspectives that differ from their own.

Here we hypothesize that these conflicting views about how individuals learn from each other can be reconciled by considering the type of characteristic that makes someone (dis-)similar.

When characteristics do not involve a strong group identity and point to potentially valuable differences in perspectives, individuals are more likely to integrate the unfamiliar approaches of dissimilar others. But when individuals strongly identify with their ingroup and have negative associations about an outgroup, unfamiliar behavior from outgroup members is disregarded and individuals are more likely to learn from similar others.

Experimental design

We test these predictions in a preregistered experiment where participants can observe the behavior of a demonstrator. The experiment involves a controlled comparison of two aspects: First, a comparison of learning from a dissimilar versus similar demonstrator. Second, a comparison between situations where dissimilarity relates to either a characteristic that could suggest a difference in perspectives, or one that activates a sense of group identity. Differences in perspective were operationalized using a cognitive style test specifically developed for this study, while group identity was based on participants' self-reported political ideology. We implement both aspects—the (dis-)similarity between learner and demonstrator, and the type of (dis-)similarity (cognitive style vs. political identity)—in a 2x2 experimental design with 859 US participants recruited via Prolific and as described below (see Methods for further details).

Approach-avoid task

The study implemented an approach-avoid task inspired by the experiment of Rich & Gureckis [24]. Participants viewed pictures of fictive creatures we called 'aliens', one alien at a time. Their task was to find out which aliens were 'friendly' and which aliens were 'hostile', and to 'approach' friendly aliens and 'avoid' hostile aliens. If a participant clicked the approach button when viewing a friendly alien, they were rewarded 2ct. If they approached a hostile alien, 6ct

were subtracted from their bonus payment. If they avoided an alien, their bonus remained unchanged (Fig. 1A, left side). When a participant approached an alien, they learned immediately whether the alien was friendly or hostile. We did not reveal if an alien was friendly or hostile if participants avoided an alien.

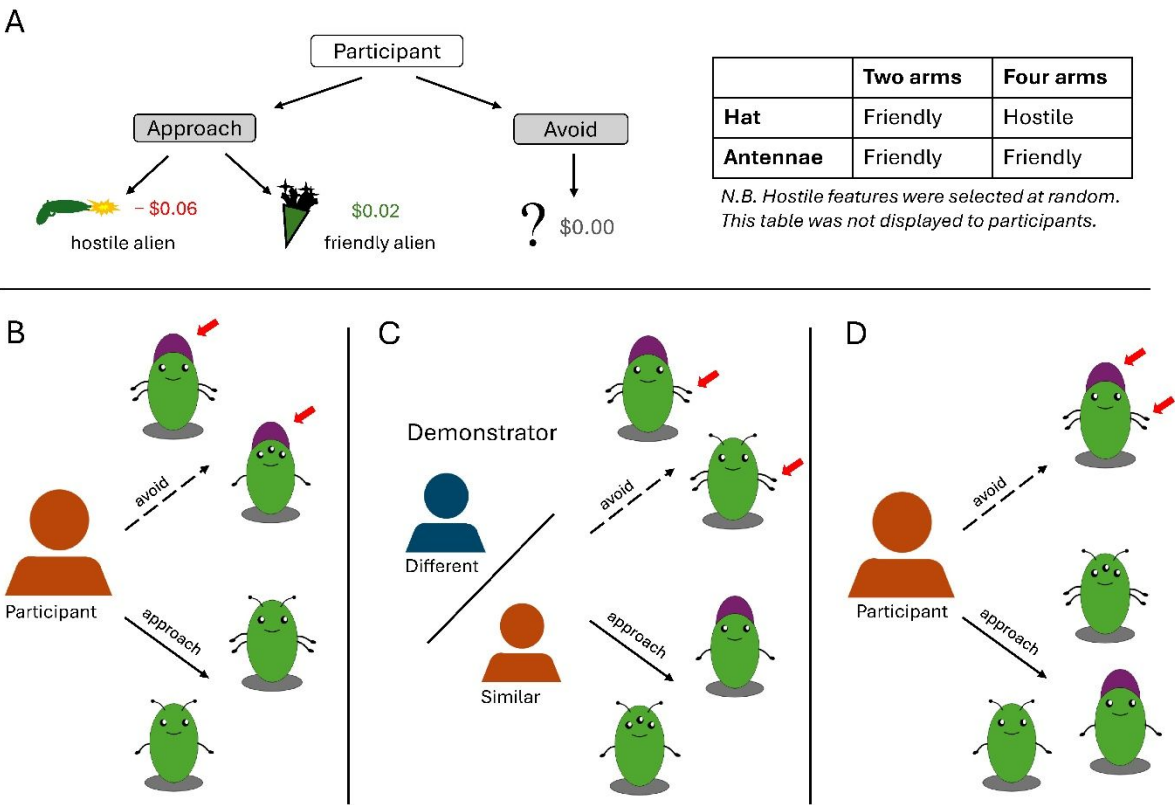


Fig. 1. Task and procedure. We instructed participants to ‘approach friendly aliens and avoid hostile ones’. Approaching friendly (hostile) aliens resulted in financial reward (punishment), avoiding aliens did not affect payment. Aliens were only hostile if two features were present (A). Participants developed behaviors where they avoided all aliens where the first hostile feature was present (B). We introduced a demonstrator with different or similar characteristics. Demonstrator behavior remained constant across conditions, avoiding all aliens with the second hostile feature and approaching aliens if this feature was not present (C). Through close observation, participants could learn to combine behaviors, optimizing their approach and only avoiding aliens with both features (D).

Aliens differed along three binary dimensions (antennae or hat, two or four arms, three or two eyes). One dimension was selected as irrelevant. The other two dimensions made an alien hostile if a combination of two specific features was present, for example, four arms and a hat; otherwise, the alien was friendly (Fig. 1A, right side). Which two features made an alien hostile was selected randomly for each participant, but it remained unchanged over time for a participant. Through trial-and-error, participants could learn which features made an alien friendly or hostile.

Learning traps

At the beginning of the experiment, participants individually made approach/avoid decisions on a number of aliens. During this ‘individual learning phase’, participants fell into ‘learning traps’ [24]: that is, they identified one feature that had to be present for an alien to be hostile but failed to identify the second feature (Fig. 1B). In such a learning trap, individuals avoid all hostile aliens (where both features are present) but also avoid friendly aliens where one feature is present but the other is not. Because participants do not learn whether aliens are friendly or hostile when they avoid them, they continue to avoid friendly aliens under the assumption that they are hostile. For this reason, learning traps represent behavior that is not optimized but stable [25, 26].

Experimental conditions

After the individual learning phase, participants indicated their political ideology (liberal / conservative) and completed a ‘cognitive style test’ (see Methods for details). This was followed by an ‘exposure phase’ where participants could see, when making an approach/avoid decision, what choice a demonstrator made when viewing the same alien. Depending on the experimental

condition, this demonstrator was presented as similar or dissimilar to the participant in terms of political ideology or cognitive style (Fig. 2). While participants were made to believe that they were observing the behavior of an earlier participant, demonstrator behavior was computer-simulated, as participants learned later in a debriefing (see Methods, ‘Debriefing’).

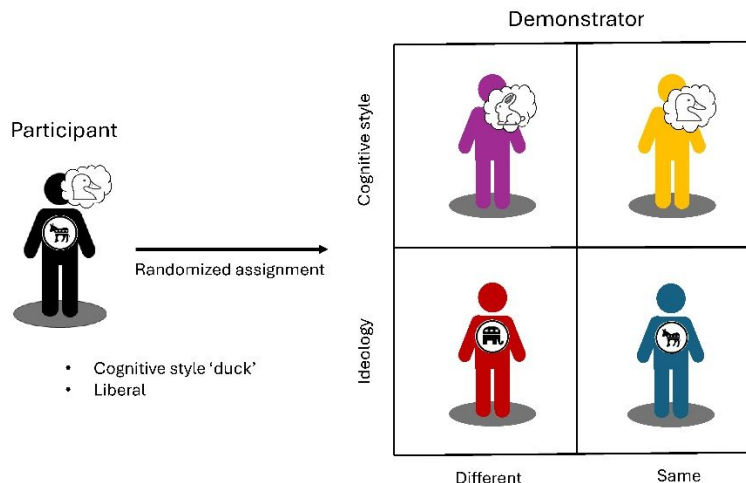


Fig. 2. Experimental conditions. Participants completed a fictional ‘cognitive style test’ and self-reported their ideology (left side). They were then matched with one of the four different demonstrator types (right side), representing experimental treatments.

Demonstrator behavior was chosen such that it was consistent with a different learning trap, i.e. the demonstrator avoided all aliens with the second hostile feature and approached aliens otherwise. In following a different learning trap, demonstrator behavior was complementary to participant behavior because it was suboptimal yet valuable, and contained both misleading and useful social information: For example, if a participant avoided all aliens with a hat, the demonstrator avoided all aliens with four arms (cf. Fig 1C). This meant that the demonstrator avoided some aliens the participant had learned to identify as friendly (those with four arms and antennae). However, it also meant that the demonstrator approached friendly aliens the

participant avoided because the participant wrongly assumed them to be hostile (those with a hat and two arms). Demonstrator behavior was the same across experimental conditions, and only the demonstrator's characteristics were varied.

The fact that the demonstrator avoided some aliens that the participant knew to be friendly made it easy to brush off demonstrator behavior as misguided. However, if a participant paid close attention and combined their own approach with that of the demonstrator, the participant could optimize: i.e., learn that some of the aliens they originally avoided were friendly. We measured whether participants integrated approaches, and thereby optimized their solution, in a final 'assessment phase', where participants made additional approach-avoid decisions individually without social information (Fig. 1D).

Results

Pooling across experimental conditions, Figure 3 presents an overview of participants' behavior in our pre-registered analysis (see Methods). 859 participants fell into a learning trap during the individual learning phase, i.e., they identified one feature that indicates hostility and began avoiding all aliens with that feature and approaching aliens otherwise. 495 (57.6 %) of them remained with their approach until the end of the experiment and were not influenced by the demonstrator. During the exposure phase, between 6.9 percent and 8.4 percent of participants switched to the demonstrator's approach. This form of social learning, where individuals simply reproduce the demonstrator's behavior, shrunk back to 2.7 percent in the assessment phase. Optimization—i.e., behavior where participants integrated their own approach with the approach of the demonstrator—appeared among 12.1 percent of participants at the start of the exposure phase and rose steadily to 30.7 percent at the end of the experiment. It persisted after demonstrator behavior was made unavailable during the assessment phase (block 10 onwards),

suggesting that participants understood the two-feature rule that made aliens hostile and were able to pursue it after the demonstrator had left.

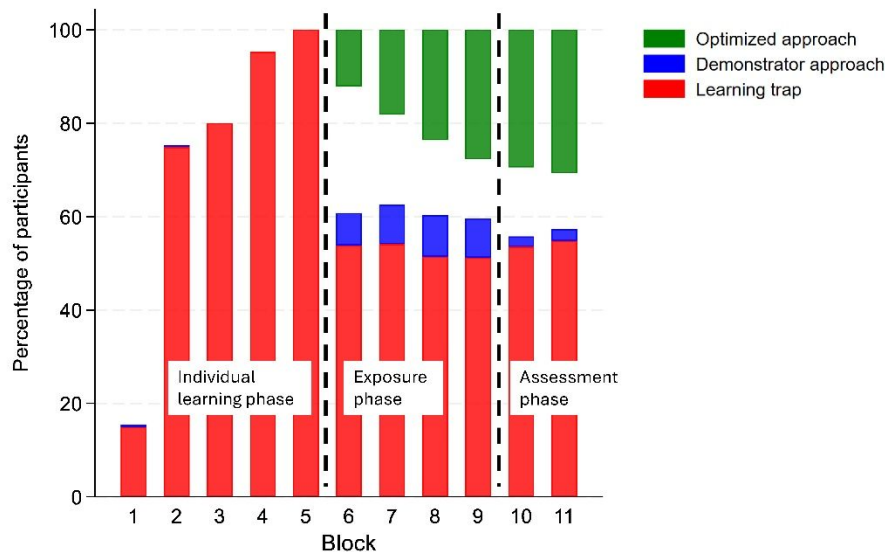


Fig. 3. Overview of participant behavior. Participants fell into learning traps during the individual learning phase. Optimization started shortly after the introduction of a demonstrator in the exposure phase. In the assessment phase, demonstrator copying diminished when participants made decisions without access to the demonstrator's decisions, while optimized approaches persisted.

As preregistered, we compare optimization after exposure to a similar versus dissimilar demonstrator, and separately for the cognitive style conditions and the ideology conditions. In line with our expectations, more participants optimized after exposure to a cognitively different demonstrator (36.7 %) than after observing a cognitively similar demonstrator (25.1 %; $p = .006$, $N = 426$). It thus appears that an indication of cognitive dissimilarity tipped participant behavior towards close observation and subsequent optimization. Ideological (dis-)similarity, on the other hand, did not significantly affect optimization rates (similar 32.5 %; dissimilar 28.3 %; $p = .41$, $N = 433$).

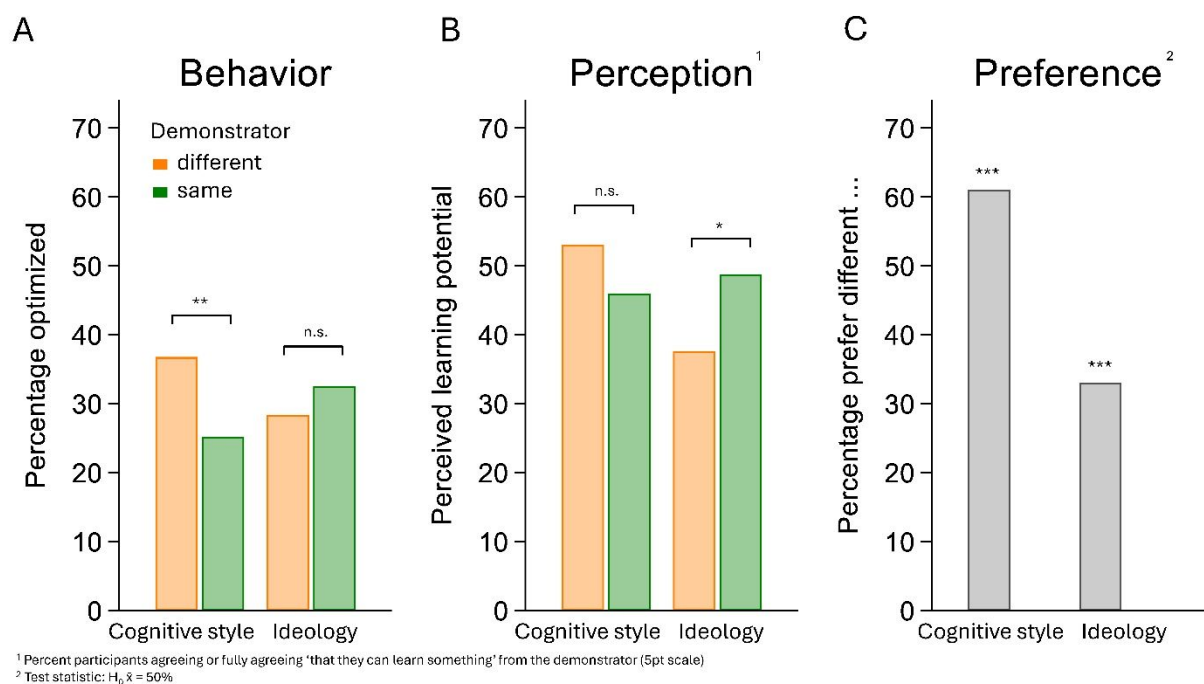


Fig. 4. Dissimilarity-driven learning arises from perceived cognitive differences but not from ideological differences. Participants optimize their solution more often after being introduced to a cognitively different demonstrator (A). They also attribute greater learning potential to ideologically similar peers (B) and prefer learning from cognitively different and ideologically similar demonstrators (C). Two-sided randomization tests: *n.s.* $p > 0.05$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

To gain more insight into how participants learned from (dis-)similar others, we implemented two additional measures. First, after having observed the demonstrator’s decisions for 8 aliens, participants indicated their agreement with the statement ‘I think that I can learn something from the earlier participant’ [i.e., the demonstrator]. If cognitively dissimilar and ideologically similar demonstrators are perceived as more valuable sources, more participants would agree or fully agree they can learn from them. Indeed, as Figure 4B shows, a greater proportion of participants (fully) agreed that they could learn from a demonstrator with a different rather than a similar cognitive style (53.0 versus 46.0 percent), but this difference falls short of statistical significance

($p = .168$, $N = 426$). However, and as expected, participants more often (fully) agreed that they can learn from the demonstrator if they were exposed to an ideologically similar one as opposed to an ideologically dissimilar one (48.7 % versus 37.6 %, $p = .025$, $N = 433$). As demonstrator behavior was identical across conditions, this confirms that ideological similarity boosts impressions of learning potential.

Second, before exposing participants to the demonstrator, we asked participants whether they preferred ‘to be matched with someone with the same cognitive style or a different cognitive style’, and someone ‘with the same ideology or a different ideology’. As illustrated in Figure 4C, the majority of participants (60.7 %) preferred a demonstrator with a different cognitive style over one with the same style (H_0 : Preference not different from chance, i.e. $\bar{x} = 50$ %; $p < .001$, $N = 859$). For political ideology, the pattern reversed: Only 33.4 percent preferred a demonstrator from the opposite ideological camp ($p < .001$, $N = 859$), suggesting that participants projected greater value onto learning from cognitively different others and ideologically similar peers.

Further analysis shows that the higher rate of optimization in the cognitive dissimilarity condition did not depend on whether participants’ preferences for a (dis-)similar demonstrator were met. This is important because, due to random assignment, more participants in the cognitive dissimilarity condition had their preferences fulfilled than in the cognitive similarity condition (Figure 4C). This could have led to a larger number of disappointed participants in the cognitive similarity condition, spuriously causing lower rates of optimization. A regression analysis reported in the SI, Section 1, reveals no significant effect for met preferences, while the effect of cognitive similarity persists. This shows that cognitive dissimilarity fostered optimization, irrespectively of whether participants desired it. Similarly, higher agreement to perceived learning potential in the ideological similarity condition was not affected by whether

1 participants' preferences were met, giving no indication that met preferences confounded
2 treatment effects.

3 **Discussion**

4 Under which conditions do individuals combine their suboptimal approach with that of someone
5 else who takes a different but likewise imperfect approach? The results of our experiment show
6 that, as hypothesized, optimization increases when demonstrator characteristics point towards
7 differences in perspectives, whereas social group identity did not affect whether participants
8 optimized. Our findings identify dissimilarity to foster social learning, but only if this
9 dissimilarity leads to expectations of learning potential. In contrast to empirical literature
10 showing that people tend to adopt behaviors from similar others [12, 13], our findings present a
11 more nuanced view of human learning. Rather than merely following ingroup members, humans
12 are capable of valuing dissimilar others more highly when they expect that someone might have
13 a different perspective that is complementary to their own.

14 Focusing on a task where combining different approaches can help individuals find the optimal
15 solution, our research sheds light on a puzzle in the literature. On the one hand, our findings
16 resonate with those of Phillips [22] and Levine et al. [27], suggesting that visible differences
17 between individuals can enhance the performance of groups. But they also connect to work that
18 highlights diversity's role as a 'double-edged sword' [28, 29]: Diverse populations perform
19 better because of a greater wealth of perspectives and skills [30, 31] but identity diversity can
20 make it challenging to fully use their potential [32, 33]. In line with this, our study showed that
21 individuals disfavored ideological difference and attributed lower learning potential to
22 ideologically different peers.

1 To overcome the challenges associated with different identities, previous research has explored
2 ways to build trust and understanding among dissimilar peers. One possibility involves
3 emphasizing ‘deep-level’ similarities, such as shared interests and values [22, 34]. However, the
4 fact that in our experiment, participants optimized less often when introduced to a ‘cognitively
5 similar’ demonstrator suggests that highlighting deep-level similarity can be counter-productive.
6 Resonating with cognitive research on curiosity-driven learning [35], our findings imply that
7 practitioners should not try to emphasize similarity at all costs but make those differences salient
8 that enable individuals to expect novelty and learning potential from each other.

9 A question that remains is why in our study, perceived cognitive differences enhanced learning
10 while ideological similarity did not. We attribute this to the fact that our experimental design
11 excluded factors that would otherwise amplify similarity bias, such as coordination advantages,
12 peer sanctioning, or a task where the viability of different approaches may be related to the social
13 categories someone belongs to [36, 37]. We do not exclude the possibility that social
14 psychological causes of identity can have a substantial impact on social learning. However, these
15 causes may instead be more applicable to settings where incentives to learn from unfamiliar
16 behavior are less strong [38], such as the exchange of opinions in online social networks [39].

17 Lastly, while research on the evolution of culture has identified reasons for why technological
18 boundaries tend to coincide with social boundaries [15, 40], our study uncovers a novel
19 mechanism that can help understand how innovative practices sometimes do emerge from
20 interactions across groups. On a larger scale, our results also link to research on the innovation
21 potential of societies [7], pointing not only to the importance of a diversity of skills and ideas,
22 but also to individual’s willingness to engage with each others’ perspectives [41]. We hope that

our results will stimulate new avenues in research, such as integrating dissimilarity-driven learning into agent-based models of cultural evolution and experimental studies.

Methods

Participant recruitment

We recruited 1600 US participants by posting a targeted advertisement on the online crowd working platform Prolific. To maximize reliability of political ideology, we only invited participants with the following criteria: they either had to have voted for Donald Trump in the 2020 US national election, identified as conservative and considered themselves republican (25 % of the sample, or 400 participants), or they had to have voted for Joe Biden, identified as liberal, and considered themselves democrats (75 %, 1200 participants). Ideological leaning was checked and verified during the experiment (see section Data quality). The study was advertised as a ‘Task about pattern recognition and learning’. Upon arrival at our experimental platform, participants received instructions and gave informed consent.

Participants were compensated with \$2.00 upon successful completion, plus a performance bonus between \$0.00 and \$1.20. The study was approved by the ethics committees of XXXX and the review board of XXXX. Data were collected between December 2, 2024 and January 10, 2025.

Individual learning phase

Throughout the experiment, participants made approach/avoid decisions on maximally eleven ‘blocks’ of eight aliens each. The individual learning phase consisted of the first five blocks,

with the first two blocks having been designed to lead participants into a specific learning trap: In these blocks, we excluded aliens where one hostile feature (“feature A”) was present and the other one (“feature B”) was absent. Because of this, participants were exposed to a stream of experiences where aliens were always hostile when feature A was present. Therefore, participants quickly arrived at the impression that only feature A mattered.

After block 2, we presented aliens such that each block featured all 8 different aliens (i.e., feature combinations). We classified behavior as consistent with a learning trap when a participant avoided all aliens with feature A and otherwise approached aliens for 8 out of 8 decisions within one block. If a subject’s behavior was consistent with a learning trap in block three or four of the individual learning phase, they immediately continued to block six. If their behavior was not yet consistent with a learning trap, they continued making approach/avoid decisions in the absence of social information for up to a fifth block.

Exposure phase

At the beginning of the exposure phase, each participant was told that they would be ‘matched with an earlier participant’ (the demonstrator) and shown ‘what decisions the earlier participant made after encountering the same aliens as you’. They then indicated their political ideology on a standard 7-point self-identification question ranging from ‘extremely conservative’ to ‘extremely liberal’[42]. Participants also completed a fictional ‘cognitive style test’ that presented them with an ambiguous image of the rabbit-duck illusion and we asked them whether they first recognized the rabbit or the duck. We informed participants that what they first saw in the image determined their ‘cognitive style’ and that cognitive style was ‘related to how people perceive patterns in the

1 alien task'. Ideology (conservative/liberal) and cognitive style (duck/rabbit) was used to let the
2 demonstrator appear as similar or dissimilar to the participant.

3 Participants subsequently continued with blocks 6 to 9, during which participants could see,
4 when making an approach/avoid decision, what choice a computer-simulated demonstrator
5 made. For each alien a participant encountered, we presented the decision of the demonstrator,
6 along with a graphical representation of their own and the demonstrators' characteristics.

7 **Assessment phase**

8 Finally, in the 'assessment phase' (blocks 10 and 11), we measured whether a participant had
9 learned to approach all friendly aliens and avoid all hostile ones in the absence of social
10 information. We classified behavior as optimized if a participant identified 8 out of 8 aliens in
11 one block correctly. Participants could finish the experiment early if they optimized their
12 approach during the exposure phase and from block 8 onward already, in which case they were
13 directly routed to a study completion page informing them about their payment.

14 **Debriefing**

15 After the study, a debriefing informed participants that the demonstrator was not a real person
16 and that 'cognitive style' was a fictional concept. We chose a computer-simulated demonstrator
17 over a human 'earlier participant' so that the demonstrator consistently followed a different
18 learning trap than the one of the participant, across experimental conditions and independent of
19 their characteristics.

20 **Sample of analysis**

We aimed to assess when individuals combine their own suboptimal behavior with that of someone else. This requires that individuals developed a suboptimal solution on their own in the first place, which was not the case for all participants. As pre-registered, we therefore restricted the final sample of analysis to those participants who had identified a suboptimal solution before they were exposed to information about the demonstrator's behavior. That is, for participants to be included in our sample, they had to have developed the specific learning trap we designed for them prior to block six: they avoided all aliens with the first of the two relevant features that made an alien hostile ("feature A") and otherwise approached all aliens for 8 out of 8 decisions within one block. This was the case for 859 out of 1600 participants recruited for the task. Of the remaining 741 participants, 180 optimized early, meaning that they identified both relevant features before being exposed to the demonstrator. 77 participants developed an 'unintended' learning trap, i.e., they identified the second hostile feature instead of the first, which meant that participant behavior was identical with demonstrator behavior, rendering exposure to the latter redundant. 471 participants followed no discernible approach, and 13 were excluded because they failed to verify their pre-screened ideology correctly, tried to participate twice, or had implausible completion times.

In the final sample of analysis, average completion time was 10 minutes and 55 seconds.

Participants made decisions on 68 aliens on average and were correct in 76 percent of their choices. With an average bonus payment of \$0.66 and \$2.00 compensation for successful completion, mean hourly compensation was \$14.63. The median age in the sample was 38 years and 57 percent identified as female. Participant demographics in the final sample were statistically indiscernible from the 741 individuals who were not included in it.

Although not the focus of our study, we investigated if participants who did not develop a learning trap on their own also exhibited similar learning from a (dis-)similar demonstrator as the participants in our main sample (SI Section 3). Results show that participants preferred ideologically similar demonstrators but otherwise no effect of cognitive or ideological (dis-)similarity. Reasons for this finding are discussed *ibidem*.

Data quality

We undertook the following steps to ensure high data quality. First, we only recruited online participants with an approval rating above 99 percent, meaning that they had to have completed nearly all their previous tasks on Prolific successfully. Second, we included timers on all instruction screens to avoid that participants did not read them carefully. Third, we verified participants' political ideology on our experimental platform, excluding individuals whose pre-screened ideology did not match their ideology when we asked them again. Fourth, we included an attention check during the experiment, asking participants in block seven if they could recall the decision of the demonstrator to approach or avoid the alien they were previously exposed to.

In line with our preregistration, the final sample of analysis also included the 14.4 percent of participants who failed this attention check. This was done to exclude the possibility that we falsely excluded participants who executed their task carefully but had decided to disregard demonstrator behavior either because of their characteristics, or because they were completely immersed in their learning trap. Robustness analyses presented in SI Section 2 show that the main results stay similar and retain their significance at conventional levels if participants who failed the attention check are excluded.

1 **Data, materials and software availability**

2 Pre-registration, data replication package, screenshots of instructions provided to participants,
3 and code for executing this experiment available at
4 https://osf.io/fjtac/?view_only=d1e9be0558b9443cac7490cfc9b9e985.

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